Draft version October 28, 2026
Typeset using LaTeX **twocolumn** style in AASTeX7.0.2

Extended Lyα Nebula Surrounding an Ultra-Massive Quiescent Galaxy at $z \sim 5$: Implications for Local Quenching Mechanisms*

Stephanie M. Urbano Stawinski[1] and friends[2]

[1]*Department of Physics & Astronomy, University of California Santa Barbara, Santa Barbara, CA 93106, USA*
[2]*Institution for friends*

ABSTRACT

1. INTRODUCTION

2. DATA

2.1. *Previous Datasets*

In March and May 2023, we observed Ultra-Massive galaxy candidate (UMG-28740) in the Extended Groth Strip (EGS) with Keck/DEIMOS **?**. It was originally selected to be massive ($M_\star > 10^{11} M_\odot$) and high-redshift ($z_{\rm phot} > 4$) from the multi-wavelength photometric catalog from the Cosmic Assembly Near-Infrared Deep Extragalactic Legacy Survey (CANDELS, **??**). We detected strong Lyα emission at $z_{\rm Ly\alpha} = 4.8947$ that had an extended profile and large equivalent width (EW), inconsistent with our best models of AGN in the joint photometric and spectroscopic SED.

In March 2024, this galaxy was observed with *JWST*/NIRSpec prism and grating (G395M) through the cycle 2 RUBIES program (GO-4233, PIs A. de Graaff and G. Brammer). These observations confirmed the quiescence of this system and placed a firm redshift of the galaxy via Hα at $z_{\rm H\alpha} = 4.8976$.

This galaxy also sits in a known overdense region in the EGS at $z \sim 5$, first identified by . This region is XX, and includes a mix of spectroscopically confirmed and photometric members .

2.2. *KCRM Observations*

In order to explore the origin of Lyα in UMG28740, we observed the region in April 2026 with the Keck Cosmic Reionization Mapper (KCRM) on Keck II. To maximize the field of view (FOV) we employed the large IFU slicer ($33'' \times 20''$) with the medium grating RM1 at a central wavelength of XX Å. We offset the center of the IFU to capture more protocluster members near UMG28740. For optimal sky subtraction, we use a 3-point dither with width of $0.7''$. We also took standard calibration (bias, flats, arcs) and observe a standard for robust flux calibration.

2.3. *Data Reduction*

The main data reduction and sky subtraction was first performed using KCWI Data Reduction Pipeline (DRP,). We use the Python-based code KCWIKit to create custom cosmic ray masks for each exposure in the KCRM data. We also use KCWIKit to produce the final stacked cube of XX exposures totaling XX sec. Astrometry corrections were performed prior to the stacking by anchoring to the *JWST*/NIRCam-based RA/Dec of bright low-z interlopers within the FOV in each exposure found in the COSMOS-Web catalog .

The final Lyα line maps were generated by first measuring the continuum level blueward of the observed-frame Lyα emission at the systematic redshift of the UMG. The spectra in each pixel was then subtracted by the continuum level, leaving a line map free of low-z continuum sources. Noise was estimated directly from the variance cube produced by the KCWIKit stacking procedure, which propagates per-exposure variance (read noise, Poisson statistics, flat-fielding uncertainty) through the combination of individual exposures.

* The data presented herein were obtained at the W. M. Keck Observatory, which is operated as a scientific partnership among the California Institute of Technology, the University of California and the National Aeronautics and Space Administration. The Observatory was made possible by the generous financial support of the W. M. Keck Foundation.

3. RESULTS
3.1. *Extent of Lyα Nebula*
3.2. *Velocity Map*
3.3. *Nebula within an Overdense Environment*
4. DISCUSSION
4.1. *Powering Source of Lyα Nebula*
4.2. *Lyα Profile*
4.3. *Local Quenching Implications of Lyα Nebula around UMG28740*

ACKNOWLEDGMENTS

The authors wish to recognize and acknowledge the very significant cultural role and reverence that the summit of Maunakea has always had within the indigenous Hawaiian community. We are most fortunate to have the opportunity to conduct observations from this mountain.

APPENDIX